Case 20: Point ignition and isotropic spread

1 Verification purpose

This case verifies the coupled surface-fire and level-set rate response to a central point ignition in uniform fuel with zero wind and zero slope. Rotational symmetry supplies a diagnostic for directional bias or raster-orientation errors, while the formal decision tests the coupling between fire behavior and level-set normal velocity. The verification question is whether the required maximum spread rate is within 10 percent of 1.51 m min^{-1}; isotropic arrival geometry is reported as supporting evidence but is not yet a separate acceptance component.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the level-set field using $\partial\phi/\partial t + V_n|\nabla\phi| = 0$. With spatially uniform fuel, moisture, wind, and slope, $V_n = R_0$ is independent of azimuth. The zero contour must therefore evolve from the ignition neighborhood toward a circle. The final spread rate raster records the local normal spread rate in metres per minute because metric spread-rate output is enabled.

The surface-fire scalar supplied to the level set is the Rothermel no-wind/no-slope rate

$$R_0 = \frac{I_R\,\xi}{\rho_b\,\epsilon\,Q_{ig}}, \qquad Q_{ig} = 250 + 1116 M_f, \tag{1}$$

where reaction intensity I_R, propagating-flux ratio ξ, bulk density ρ_b, effective heating number ϵ, and fuel moisture M_f are evaluated from the generated Fuel Model 3 and moisture inputs. This equation chain follows Andrews (2018), *The Rothermel Surface Fire Spread Model and Associated Developments*; the Guide supplies the configured known-response value.

3 Derivation of expected behavior

For constant R_0, the front radius is $r(t) = r(0) + R_0 t$. The independently tabulated Fuel Model 3 response for the configured moisture and no-wind/no-slope condition is $R_0 = 1.51$ m min^{-1}. Rotational symmetry is an invariant; the quantitative decision uses the rate because it directly tests the fire-behavior value passed to propagation.

The independent no-wind/no-slope reference is calculated from

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \qquad Q_{ig} = 250 + 1116 M_f,$$

using generated Fuel Model 3 and $(M_{1h}, M_{10h}, M_{100h}, M_{lh}, M_{lw}) = (0.06, 0.07, 0.08, 0.60, 0.90)$. The independent calculation gives $I_R = 549.174$ kW m^{-2} and $R_0 = 1.51183$ m min^{-1}, reported to the Guide precision as $R_{\text{ref}} = 1.51$ m min^{-1} ($1.51183/0.3048 = 4.960$ ft min^{-1}). The decision metric is therefore

$$e_R = \frac{|R_{\max} - 1.51|}{1.51},$$

where $R_{\max}$ is the maximum finite final spread rate value, and the substituted acceptance test is $e_R \leq 0.10$.

4 Verification metrics and acceptance criteria

The head rate of spread (ROS) metric is the maximum finite value in the required final spread rate raster. It is selected because uniform forcing makes every mature front direction equivalent and the maximum avoids dilution by unburned nodata cells. A 10 percent relative tolerance follows the GUIDE comparison and allows raster and front-discretization effects while remaining sensitive to unit or coupling errors. The case passes only when the output exists and $|R_{\text{sim}} - 1.51|/1.51 \leq 0.10$.

5 Simulation configuration and predicted outcome

The case uses a 126×126, 30 m grid in EPSG:32635, uniform Fuel Model 3, 6 percent one-hour moisture, flat terrain, zero wind, and a central ignition. The predicted output is an approximately circular arrival-time field with a uniform mature spread rate near 1.51 m min^{-1}.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

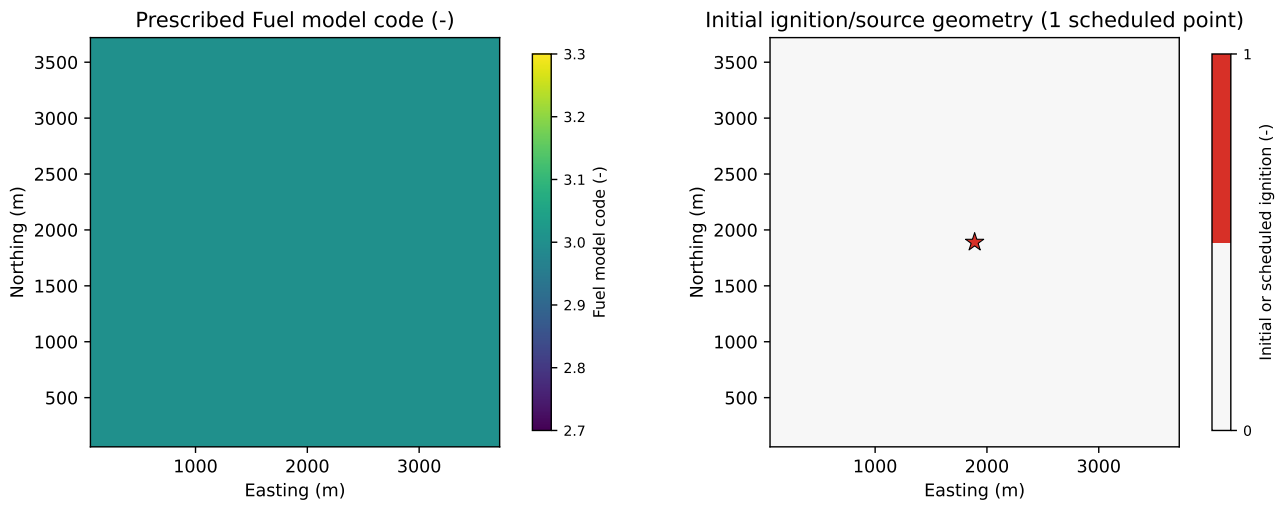


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

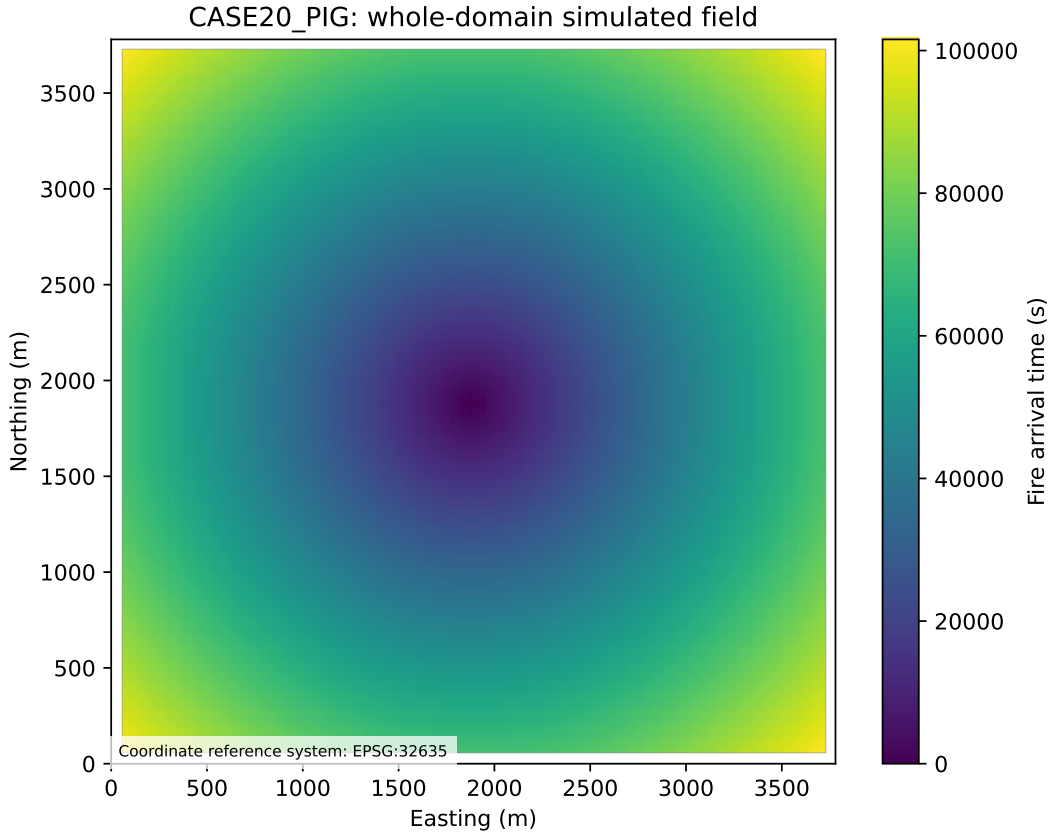


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / head ROS	1.51 m/min	1.51183 m/min	PASS

The displayed overall decision is **PASS**. When calculated verification and validation metrics exists, result analysis procedure generates every table value and status; when it is absent, the report states NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate failure indicates a fire-behavior, units, or output-selection problem. Strong directional distortion with an acceptable maximum is a diagnostic warning and must not be claimed as verified until a shape metric and tolerance are added. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

This is deterministic numerical verification for one fuel and moisture state. It does not validate Fuel Model 3 against experimental fires or quantify symmetry with a dedicated shape metric. This case provides implementation verification evidence only.